📘 PART 3 — LEVEL 3 (Advanced MCQs – JEE Main) ~130 Questions

🔶 SECTION 1: Crystal Lattices & Unit Cells (Q1–40)

1. Edge length of a cubic unit cell = 4 Å. Atomic radius of atoms = ? (Assume SC)
   A) 1 Å
   B) 2 Å
   C) 4 Å
   D) 0.5 Å
2. Density of a metal (FCC) = 8 g/cm^3, atomic mass = 64 g/mol. Calculate edge length.
   A) 3.6 Å
   B) 4 Å
   C) 4.5 Å
   D) 5 Å
3. Number of atoms in BCC unit cell = ?
   A) 2
   B) 4
   C) 8
   D) 1
4. Packing efficiency of FCC = ?
   A) 52%
   B) 68%
   C) 74%
   D) 60%
5. Atomic radius in FCC metal = 1.43 Å. Calculate edge length.
   A) 4.05 Å
   B) 3 Å
   C) 4.5 Å
   D) 5 Å
6. SC lattice has density = 5 g/cm^3, atomic mass = 50 g/mol. Find edge length.
   A) 2 Å
   B) 3 Å
   C) 4 Å
   D) 5 Å
7. A metal crystallizes in BCC structure with edge length 3 Å. Find radius of atom.
   A) 1 Å
   B) 1.5 Å
   C) 1.73 Å
   D) 2 Å
8. Coordination number in FCC = ?
   A) 6
   B) 8
   C) 12
   D) 4

9. Coordination number in BCC = ?
   A) 6
   B) 8
   C) 12
   D) 4
10. Ratio of radii of cation to anion for stable cubic lattice (CsCl type) = ?
   A) 0.414
   B) 0.732
   C) 1
   D) 0.225
11. Number of tetrahedral voids per unit cell in FCC = ?
   A) 4
   B) 6
   C) 8
   D) 2
12. Number of octahedral voids per unit cell in FCC = ?
   A) 4
   B) 6
   C) 8
   D) 12
13. Edge length of BCC unit cell = 2.87 Å. Atomic radius = ?
   A) 1.43 Å
   B) 1 Å
   C) 1.5 Å
   D) 2 Å
14. Packing efficiency of BCC = ?
   A) 68%
   B) 74%
   C) 52%
   D) 60%
15. Number of atoms in HCP unit cell = ?
   A) 2
   B) 4
   C) 6
   D) 8
16. Atomic packing factor of HCP = ?
   A) 52%
   B) 68%
   C) 74%
   D) 60%
17. SC lattice contains atoms of mass 50 g/mol, density 5 g/cm^3. Edge length = ?
   A) 3 Å
   B) 4 Å
   C) 2.5 Å
   D) 3.5 Å
18. Volume of FCC unit cell with edge length 4 Å = ?
   A) 64×10^{-24} cm^3
   B) 16×10^{-24} cm^3

C) 32×10^{-24} cm^3
D) 8×10^{-24} cm^3

19. Fractional volume occupied by atoms in FCC = ?
    A) 0.52
    B) 0.68
    C) 0.74
    D) 0.60
20. Edge length of NaCl unit cell = 5.64 Å. Calculate distance between nearest Na^+ and Cl^- ions.
    A) 2.82 Å
    B) 3 Å
    C) 2 Å
    D) 4 Å
21. CsCl lattice: distance between Cs^+ and Cl^- ions = 2.68 Å. Find radius of Cs^+ if r_Cl = 1.67 Å.
    A) 1.01 Å
    B) 1.2 Å
    C) 0.95 Å
    D) 1.5 Å
22. Volume of tetrahedral void in terms of atomic radius r = ?
    A) 0.225 r
    B) 0.414 r
    C) 0.732 r
    D) 0.155 r
23. Volume of octahedral void in terms of atomic radius r = ?
    A) 0.225 r
    B) 0.414 r
    C) 0.732 r
    D) 0.155 r
24. Fraction of voids in SC lattice = ?
    A) 0.48
    B) 0.32
    C) 0.26
    D) 0.52
25. Fraction of voids in FCC lattice = ?
    A) 0.26
    B) 0.32
    C) 0.48
    D) 0.52
26. Volume occupied by atoms in BCC = ?
    A) 0.68
    B) 0.74
    C) 0.52
    D) 0.60
27. Edge length of SC lattice = 3 Å. Atomic radius = ?
    A) 1.5 Å
    B) 2 Å
    C) 1 Å
    D) 1.43 Å

28. Atomic radius in HCP unit cell = ?
    A) $2r = a$
    B) $a = 2\sqrt{2}\,r$
    C) $a = 2r/\sqrt{3}$
    D) $a = r$
29. If density of metal doubles, volume occupied by atoms changes:
    A) Doubles
    B) Halves
    C) Same
    D) Quadruples
30. Number of atoms along body diagonal of BCC = ?
    A) 1
    B) 2
    C) 3
    D) 4
31. Number of atoms along face diagonal of FCC = ?
    A) 1
    B) 2
    C) 3
    D) 4
32. Edge length of BCC unit cell in terms of radius r:
    A) $a = 4r/\sqrt{3}$
    B) $a = 2r$
    C) $a = 2\sqrt{2}\,r$
    D) $a = r$
33. Edge length of FCC unit cell in terms of r:
    A) $a = 4r/\sqrt{3}$
    B) $a = 2r$
    C) $a = 2\sqrt{2}\,r$
    D) $a = r$
34. If radius of atoms increases, packing efficiency:
    A) Increases
    B) Decreases
    C) Remains same
    D) Zero
35. Fraction of tetrahedral voids per atom in FCC:
    A) 2
    B) 1
    C) 4
    D) 6
36. Fraction of octahedral voids per atom in FCC:
    A) 2
    B) 1
    C) 3
    D) 4
37. Volume of unit cell = 64 $Å^3$. Edge length = ?
    A) 4 Å
    B) 8 Å

C) 16 Å
D) 32 Å

38. Density formula: ρ = ?
    A) $ZM/(a^3 N\_A)$
    B) $a^3/(ZM N\_A)$
    C) $M/(Z a^3 N\_A)$
    D) $N\_A/(Z M a^3)$
39. Number of atoms in FCC = ?
    A) 4
    B) 2
    C) 1
    D) 8
40. Number of atoms in BCC = ?
    A) 2
    B) 1
    C) 4
    D) 8